MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VII SEMESTER B.TECH.

MAKE UP END SEMESTER EXAMINATION (JAN 2024)

SUBJECT: AUGMENTED AND VIRTUAL REALITY (PE-III) – CSE 4051

Duration: 3 Hours Date: 05/01/2024 MAX.MARKS: 50

Note:

1. Missing data may be assumed suitably.2.30 PM

Q.No	Questions	Marks	CO /CLO	AHEP LO	Blooms Taxonomy level
1a)	Detail the technology behind the data and power glove with a neat sketch.	3	1	1,3	2
1b)	Differentiate between virtual and augmented reality in terms of scope and illustration.	3	1	1,3	2
1c)	Which businesses are using or developing AR technology? Could you summarize a few recent developments in augmented and virtual reality?	4	1	1,3	3
2a)	Why does augmented reality typically improve our visual sense of the real world?	3	2	3	4
2b)	When there isn't a tangible object with all the necessary characteristics to convey the sense of touch, providing a realistic sensory impression might be challenging. Justify?	3	2	3	4
2c)	Write down the various implications for stereo viewing devices. Mention how auto-stereoscopic displays function.	4	2	3	2
3a)	Examine the three crucial terms related to object alignment and measurement.	3	3	2	4
3b)	Categorize the marker tracker pipeline's five stages.	3	3	2	4
3c)	Analyze the homography-based POSE estimation method. How is pose refinement's reprojection error minimized?	4	3	2	4

4a)	Examine the methods used to synchronize the haptic and visual pipelines for both virtual and augmented reality modalities.	3	4	2,3	4
4b)	Recognize the issues with displays that aren't synchronized between augmented and virtual reality modes?	3	4	2,3	4
4c)	What is the hierarchy of the three phases in the graphics rendering pipeline? Analyze every phase in depth.	4	4	2,3	4
5a)	Find and discuss a model that provides a detailed mathematical description of shapes.	3	5	1,3,5	4
5b)	Outline the prominent features of Polhemus 3D scanners. How do you compare Deltasphere and Polhemus scanners?	3	5	1,3,5	2
5c)	Compare and contrast the processes involved regarding the geometric modeling techniques used in Surface vs. Solid modeling. List the benefits and drawbacks of each modeling technique.	4	5	1,3,5	3

Abbreviations:

M – Marks

CO/CLO– Course Outcome (NBA)/Course Learning Outcome (IET).

CLO – Course Learning Outcome as per AHEP 4

BT - Blooms Taxonomy Level